Inequalities Error Analysis — ANSWER KEY

Find the mistake. Explain it. Fix it.

Suggested responses; accept equivalent mathematical reasoning.

1. Dividing by a negative

Mistake:

The student divided by −3 but did not reverse >.

Correct work:

$-3x > 18$; $x < -6$

2. Distributing a negative

Mistake:

$-2(-5)$ equals +10, not −10.

Correct work:

$-2x + 10 \leq 14$; $-2x \leq 4$; $x \geq -2$

3. Combining unlike terms

Mistake:

2x and 11 are unlike terms; you cannot combine them into 13x.

Correct work:

$4x + 3 < 2x + 11$; $2x < 8$; $x < 4$

4. Compound inequality and negatives

Mistake:

Dividing by a negative reverses BOTH comparison signs; then reorder the bounds.

Correct work:

$4 > x \geq -2$; $-2 \leq x < 4$

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5. AND versus OR

Mistake:

The intersection of the two conditions is the interval between -2 and 5, not the outside rays.

Correct work:

$-2 < x < 5$

6. Open and closed endpoints

Mistake:

Since $\geq$ includes -1, the endpoint must be closed.

Correct work:

Closed circle at -1; shade right.

7. Flipping the inequality sign

Mistake:

Dividing by -2 reverses $<$ to $>$.

Correct work:

$-2x < 6$; $x > -3$

8. OR: forgetting a branch

Mistake:

The student solved only the first inequality and discarded the second condition.

Correct work:

$x < -2$ OR $x \geq 4$

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9. Recognizing no solution

Mistake:

6 > 9 is false, so NO number makes the original statement true.

Correct work:

6 > 9 is false; no solution

10. Writing a real-world inequality

Mistake:

“At most” means ≤, not ≥. Whole tickets and a nonnegative count are required.

Correct work:

12t ≤ 50; t ≤ 50/12; 0 ≤ t ≤ 4 (whole tickets).